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Determination of the Possible Target Genes of Hepatoma-derived Growth Factor in Hepatoma Cells

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Abstract. *Background/Aim: We identified a new growth factor, hepatoma-derived growth factor (HDGF), which is a presumed growth-stimulating factor of hepatocellular carcinoma (HCC). Recently, we identified two microRNAs (miR-6072 and miR-3137) induced by HDGF, which were also found to be associated with the prognosis of HCC patients. This study aimed to identify the target genes of these HDGF-related microRNAs. Materials and Methods: A public database was searched for candidate target genes of HDGF-related microRNAs. Using the microarray system, the genes whose expression changed in response to HDGF administration were determined. Finally, a public cancer genomics database was searched for genes that were induced by HDGF and associated with the prognosis of HCC. Results: A total of 1,132 genes were identified as common target genes of the 2 HDGF-related microRNAs. Among these genes, a microarray system showed that the expression of 6 genes was increased (≥1.5-fold) or decreased (≤0.67-fold) after HDGF administration. Using a cancer genomics database, two of the six genes were found to be related to the prognosis of HCC. A high expression of alkylglycerone phosphate synthase (AGPS) was significantly associated with a poor survival (p=0.0025, 0.0063 and 0.0081 for the 1-, 3- and 5-year survival, respectively). A high expression of the shroom family member 4 (SHROOM4) gene was found to be significantly associated with a better survival (p=0.003, 0.0006 and 0.0006 for the 1-, 3- and 5-year survival, respectively). Conclusion: This study identified potential target genes of HDGF-related microRNAs that were associated with the prognosis of HCC.*

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Key Words: Hepatoma-derived growth factor, hepatocellular carcinoma, target genes, prognosis.

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Hepatocellular carcinoma (HCC) is one of the most common malignant diseases in the world (1, 2). Despite recent advances in drug therapy for HCC, the prognosis of HCC remains unsatisfactory (3-5), and new approaches for the treatment of HCC are still sought.

We searched for a novel growth factor that stimulated the proliferation of hepatoma cells and identified the new molecule hepatoma-derived growth factor (HDGF) (6). We previously reported that HDGF functions as a growth-stimulating factor for hepatic cells and that its high expression is associated with a poor prognosis in patients with HCC (6, 7). In addition, HDGF has been suggested to play a significant role in various kinds of malignancies (8-12).

To assess the mechanisms underlying HDGF-related cellular proliferation, we searched for microRNAs that were commonly induced in two hepatoma cell lines after HDGF stimulation and suggested that miR-6072 and miR-3137 were associated with HDGF-induced cellular proliferation (13). However, which genes are influenced by these microRNAs is unclear.

This study therefore investigated which genes are targeted by the abovementioned HDGF-related microRNAs and relate to the prognosis of HCC patients.

Materials and Methods

Cell culture and RNA extraction. In our previous study (13), we identified two microRNAs that were induced by HDGF administration and associated with the prognosis of patients with HCC using two hepatoma cell lines (HepG2 and SK-Hep1 cells). The summary of the cell culture methods was as follows: hepatoma cells were inoculated in 12-well dishes (Corning International, Tokyo, Japan) containing Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum (Sigma-Aldrich Japan, Tokyo, Japan). Twenty-four hours after the inoculation of the cells, recombinant HDGF (Abcam, Tokyo, Japan;

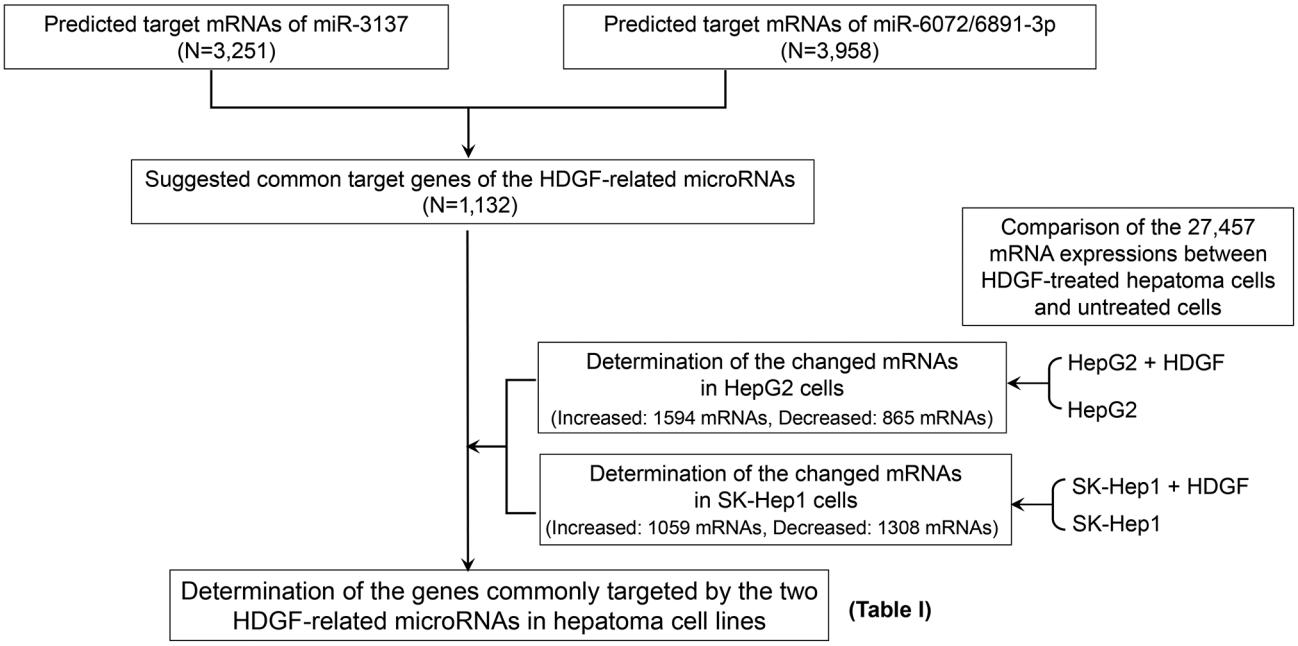


Figure 1. *Flowchart to determine the genes that were commonly targeted by the two HDGF-related microRNAs and whose expression changed in response to hepatoma-derived growth factor (HDGF). A public database was used to explore the candidate target genes of HDGF-related microRNAs. Using the microarray system with the panels of 27,457 mRNAs, the genes that were commonly targeted by the two HDGF-related microRNAs in hepatoma cell lines were then determined.*

product code: ab132259) was administered at a final concentration of 100 ng/ml in the medium.

After 48 h of HDGF stimulation, total RNA was isolated using the miRNeasy Mini Kit (Qiagen-Japan, Tokyo, Japan) according to the manufacturer's instructions. Total RNA was also collected from control cells without HDGF administration.

Identification of mRNAs suggested to be targeted by HDGF-related microRNAs (Figure 1). A public database that provided target genes of microRNAs was accessed (Target Scan Human, version 8.0) (14, 15) and the candidate target genes of HDGF-related microRNAs (miR-3137 and miR-6072) were explored. In this database, suggested target genes of miR-6072 were provided as those of miR-6072/6891-3p. Using the microarray system with panels of 27,457 mRNAs (Human Oligo chip 25k Ver. 2.10, Toray, Japan), the mRNA expression of HDGF-treated cells was compared with that of control (untreated) cells and the mRNAs that were increased (≥1.5-fold) or decreased (≤0.67-fold) after HDGF administration were determined. The SK-Hep1 hepatoma cell line was also used to search for genes whose expression changed after HDGF stimulation. In this manner, we determined the possible target genes of HDGF-related microRNAs in hepatoma cell lines (Figure 1).

Association of the mRNAs with the survival of primary liver cancer. We used public cancer genomics datasets (UCSC Xena platform) (16) to assess whether or not the determined possible target genes of the HDGF-related microRNAs were related to the prognosis of patients with liver cancer. Based on the data in the system (GDC TCGA Liver Cancer), a total of 351 patients were categorized into two groups—a high-expression group and a low-expression group—according to the median expression. The overall survival curve between the two groups was evaluated by the log-rank test. A *p*-value <0.05 was considered to represent a significant difference in the prognosis.

Table I. *Possible common target genes of the two HDGF-related microRNAs (miR-6072 and miR-3137).*

Commonly increased mRNAs
AGPS: Alkylglycerone phosphate synthase
ENGASE: Endo-beta-N-acetylglucosaminidase
NHS: Nance-Horan syndrome (congenital cataracts and dental anomalies)
RALGAPB: Ral GTPase activating protein, beta subunit (non-catalytic)
SHROOM4: Shroom family member 4
ZNF641: Zinc finger protein 641

Results

Using the public database, we identified a total of 1,132 genes suggested to be common target genes of the two HDGF-related microRNAs. Using the microarray system, we next evaluated whether or not the mRNA expression of such genes was altered in response to the administration of HDGF protein (Figure 1). Among the 1,132 genes, no gene was found to be commonly decreased (≤0.67-fold) in the two hepatoma cell lines, whereas the expression of six genes was

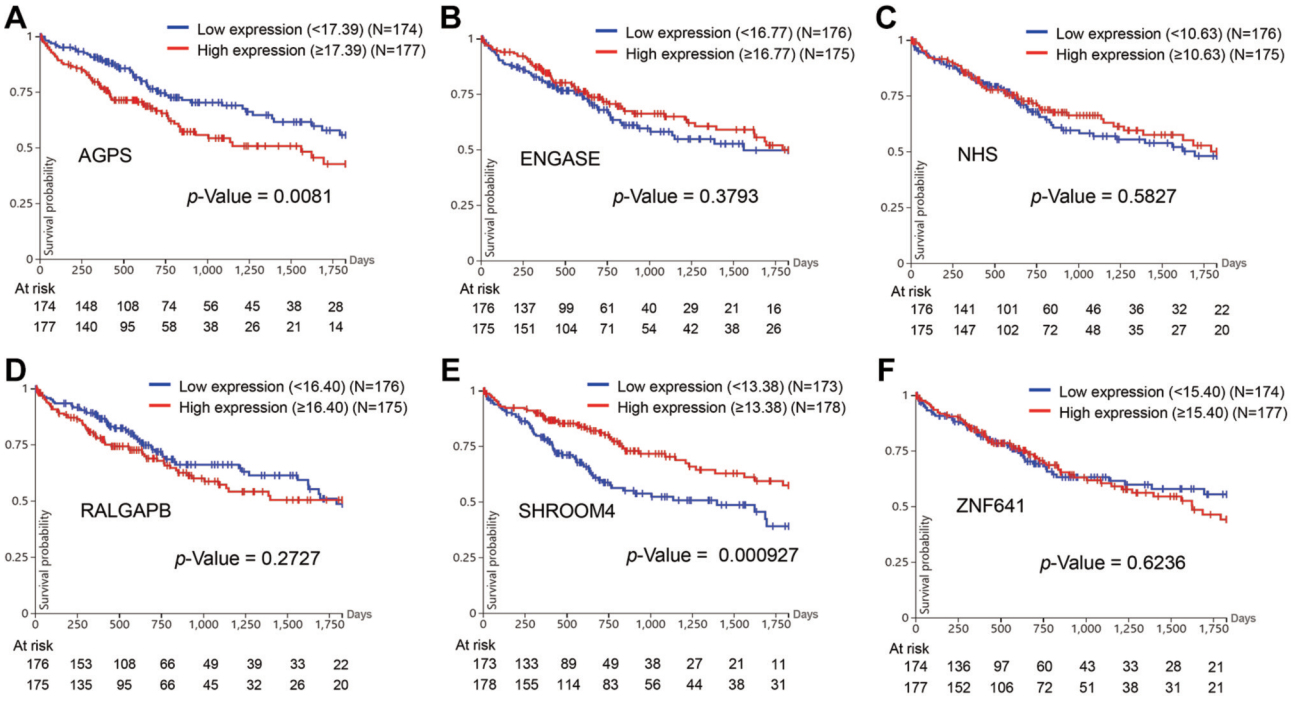


Figure 2. *Association of the six genes targeted by HDGF-related microRNAs with the five-year survival of hepatocellular carcinoma (HCC). Using the UCSC Xena platform, we investigated whether or not the expression of the six genes targeted by the HDGF-related microRNAs was related to the prognosis of patients with HCC. The Kaplan-Meier survival curves are shown. (A) Alkylglycerone phosphate synthase (AGPS), (B) endo-beta-N-acetylglucosaminidase (ENGASE), (C) Nance-Horan syndrome (NHS), (D) Ral GTPase activating protein, beta subunit, noncatalytic (RALGAPB), (E) Shroom family member 4 (SHROOM4), and (F) Zinc finger protein 641 (ZNF641).*

commonly increased (≥1.5-fold) in the two hepatoma-derived cell lines after stimulation with HDGF (Table I).

We further evaluated the association of the six candidate genes with the prognosis of patients with liver cancer. Using the cancer genomics database with 351 HCC patients, we found that a high expression of alkylglycerone phosphate synthase (*AGPS*) was significantly associated with a poor survival (p=0.0025, 0.0063 and 0.0081 for the 1-, 3- and 5-year survival, respectively). In contrast, a high expression of the shroom family member 4 (*SHROOM4*) gene was significantly associated with a favorable prognosis and better survival (p=0.003, 0.0006 and 0.0006 for the 1-, 3- and 5-year survival, respectively) (Figure 2).

Discussion

Because of advancements in antiviral treatments and lifestyle changes, the etiologies of HCC have been changing in recent years (17, 18); however, HCC is still a highly frequent malignancy, remaining a major health concern worldwide (1, 2), and research regarding the mechanisms of HCC growth is important.

We discovered the original growth factor HDGF and reported microRNAs that were induced by HDGF and associated with a poor prognosis (13). This study further explored genes that were considered to be common targets of HDGF-related microRNAs and associated with the prognosis of HCC patients.

AGPS is a protein involved in lipid synthesis, and previous studies have suggested that it affects intracellular lipid metabolism and contributes to the proliferation of malignant cells, including hepatoma cells (19-22). Our findings are consistent with these reports, and our current study is the first report to show the possible involvement of AGPS in the prognosis of patients with HCC. Recent studies have suggested the contribution of HDGF to lipogenesis-related events, including the proliferation of cancer cells (23, 24); thus, the induction of AGPS may be involved in HDGF-related metabolic changes and the growth stimulation of hepatoma cells.

A high expression of SHROOM4 was associated with an improved prognosis, although its expression was induced by HDGF stimulation. Regarding cancer cell proliferation, positive feedback systems are generally the focus of much attention. However, negative feedback systems are considered to be involved in various physiological aspects, including cancer cell proliferation (25, 26). We suspect that the induction of SHROOM4 may be a negative feedback

reaction in response to the stimulation of HDGF. However, the reason for the favorable outcome in cases with a high expression of SHROOM4 should be further investigated.

Several limitations associated with the present study warrant mention. We previously reported two HDGF-related microRNAs, and the current study focused on the common target genes of these microRNAs; however, additional studies on genes without an association with these microRNAs are also needed. Although the relationships of *AGPS* and *SHROM4* genes with the prognoses of patients with HCC have barely been reported, functional assays of the two genes in hepatoma growth have not been conducted.

In summary, this study focused on the genes targeted by HDGF-related microRNAs and identified two genes with possible associations with the prognosis of patients with HCC.

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Conflicts of Interest

The Authors declare that there are no conflicts of interest in association with this study.

Authors' Contributions

HE: Study design, data analysis, funding acquisition and the original draft writing; TN, SF, HS: performed the experiments and data analysis; HI: study design, data interpretation, review and editing the manuscript. All Authors read and edited the manuscript and approved the final version of the manuscript.

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References

1 Vogel A, Meyer T, Sapisochin G, Salem R, Saborowski A: Hepatocellular carcinoma. Lancet 400(10360): 1345-1362, 2022. DOI: 10.1016/S0140-6736(22)01200-4

2 Sung H, Ferlay J, Siegel RL, Laversanne M, Soerjomataram I, Jemal A, Bray F: Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J Clin 71(3): 209-249, 2021. DOI: 10.3322/caac.21660

3 Lee YC, Huang WT, Lee MY, Tsao CJ, Feng YH: Bevacizumab and atezolizumab for unresectable hepatocellular carcinoma: Real-world data in Taiwan-Tainan Medical Oncology Group H01 trial. In Vivo 37(1): 454-460, 2023. DOI: 10.21873/invivo.13099

4 Alawyia B, Constantinou C: Hepatocellular carcinoma: a narrative review on current knowledge and future prospects. Curr Treat Options Oncol 24(7): 711-724, 2023. DOI: 10.1007/s11864-023-01098-9

5 Chen Y, Chen Y, Wang J, Hung C: Efficacy and safety of lenvatinib after progression on first-line atezolizumab plus bevacizumab treatment in advanced hepatocellular carcinoma patients. Anticancer Res 43(3): 1377-1384, 2023. DOI: 10.21873/anticanres.16286

6 Enomoto H, Nakamura H, Nishikawa H, Nishiguchi S, Iijima H: Hepatoma-derived growth factor: an overview and its role as a potential therapeutic target molecule for digestive malignancies. Int J Mol Sci 21(12): 4216, 2020. DOI: 10.3390/ijms21124216

7 Enomoto H, Nakamura H, Liu W, Iwata Y, Nishikawa H, Takata R, Yoh K, Hasegawa K, Ishii A, Takashima T, Sakai Y, Aizawa N, Ikeda N, Iijima H, Nishiguchi S: Down-regulation of HDGF inhibits the growth of hepatocellular carcinoma cells *in vitro* and *in vivo*. Anticancer Res 35(12): 6475-6479, 2015.

8 Bao C, Liu K, Wang X, Ma W, Wang J, Wang C, Jia Y, Wang N, Tan B, Song Q, Cheng Y: Prognostic role of hepatoma-derived growth factor in solid tumors of Eastern Asia: a systematic review and meta-analysis. Asian Pac J Cancer Prev 16(5): 1803-1811, 2015. DOI: 10.7314/apjcp.2015.16.5.1803

9 Bao C, Wang J, Ma W, Wang X, Cheng Y: HDGF: a novel jack-of-all-trades in cancer. Future Oncol 10(16): 2675-2685, 2014. DOI: 10.2217/fon.14.194

10 Lee H, Tsai K, Lee C: Prognostic influence of cytoplasmic/nuclear hepatoma-derived growth factor in head and neck cancer. Anticancer Res 41(2): 803-810, 2021. DOI: 10.21873/anticanres.14832

11 Wang Q, Chen C, Ding Q, Zhao Y, Wang Z, Chen J, Jiang Z, Zhang Y, Xu G, Zhang J, Zhou J, Sun B, Zou X, Wang S: METTL3-mediated m^6A modification of HDGF mRNA promotes gastric cancer progression and has prognostic significance. Gut 69(7): 1193-1205, 2020. DOI: 10.1136/gutjnl-2019-319639

12 Koh HM, Hyun CL, Jang BG, Lee HJ: The relationship between hepatoma-derived growth factor and prognosis in non-small cell lung cancer: A systematic review and meta-analysis. Medicine (Baltimore) 99(51): e23837, 2020. DOI: 10.1097/MD.0000000000023837

13 Enomoto H, Nakamura H, Nishikawa H, Nishimura T, Iwata Y, Nishiguchi S, Iijima H: Hepatocellular carcinoma-associated microRNAs induced by hepatoma-derived growth factor stimulation. In Vivo 34(5): 2297-2301, 2020. DOI: 10.21873/invivo.12041

14 Huang Y, Qian M, Chu J, Chen L, Jian W, Wang G: Identification of circRNA-miRNA-mRNA network in luminal breast cancers by integrated analysis of microarray datasets. Front Mol Biosci 10: 1162259, 2023. DOI: 10.3389/fmolb.2023.1162259

15 Barut Z, Cabbar AT, Yilmaz SG, Akdeniz FT, Simsek MA, Capar B, Degertekin M, Dalan AB, Yerebakan H, Isbir T: Investigation of circulating miRNA-133, miRNA-26, and miRNA-378 as candidate biomarkers for left ventricular hypertrophy. In Vivo 35(3): 1605-1610, 2021. DOI: 10.21873/invivo.12417

16 Heyliger SO, Soliman KFA, Saulsbury MD, Reams RR: Prognostic relevance of ZNF844 and Chr 19p13.2 KRAB-Zinc finger proteins in clear cell renal carcinoma. Cancer Genomics Proteomics 19(3): 305-327, 2022. DOI: 10.21873/cgp.20322

17 Toh MR, Wong EYT, Wong SH, Ng AWT, Loo L, Chow PK, Ngeow J: Global epidemiology and genetics of hepatocellular

carcinoma. Gastroenterology 164(5): 766-782, 2023. DOI: 10.1053/j.gastro.2023.01.033
18 Enomoto H, Ueno Y, Hiasa Y, Nishikawa H, Hige S, Takikawa Y, Taniai M, Ishikawa T, Yasui K, Takaki A, Takaguchi K, Ido A, Kurosaki M, Kanto T, Nishiguchi S, Japan Etiology of Liver Cirrhosis Study Group in the 54th Annual Meeting of JSH: The transition in the etiologies of hepatocellular carcinoma-complicated liver cirrhosis in a nationwide survey of Japan. J Gastroenterol 56(2): 158-167, 2021. DOI: 10.1007/s00535-020-01748-x
19 Zhu Y, Li WM, Zhang L, Xue J, Zhao M, Yang P: Inhibitory effect of isothiocyanate derivant targeting AGPS by computer-aid drug design on proliferation of glioma and hepatic carcinoma cells. Int J Clin Exp Pathol 8(1): 812-817, 2015.
20 Zhu Y, Liu X, Yang P, Zhao M, Lv L, Zhang G, Wang Q, Zhang L: Alkylglyceronephosphate synthase (AGPS) alters lipid signaling pathways and supports chemotherapy resistance of glioma and hepatic carcinoma cell lines. Asian Pac J Cancer Prev 15(7): 3219-3226, 2014. DOI: 10.7314/apjcp.2014.15.7.3219
21 Stazi G, Battistelli C, Piano V, Mazzone R, Marrocco B, Marchese S, Louie SM, Zwergel C, Antonini L, Patsilinakos A, Ragno R, Viviano M, Sbardella G, Ciogli A, Fabrizi G, Cirilli R, Strippoli R, Marchetti A, Tripodi M, Nomura DK, Mattevi A, Mai A, Valente S: Development of alkyl glycerone phosphate synthase inhibitors: Structure-activity relationship and effects on ether lipids and epithelial-mesenchymal transition in cancer cells. Eur J Med Chem 163: 722-735, 2019. DOI: 10.1016/j.ejmech.2018.11.050
22 Hou S, Tan J, Yang B, He L, Zhu Y: Effect of alkylglycerone phosphate synthase on the expression profile of circRNAs in the human thyroid cancer cell line FRO. Oncol Lett 15(5): 7889-7899, 2018. DOI: 10.3892/ol.2018.8356
23 Min X, Wen J, Zhao L, Wang K, Li Q, Huang G, Liu J, Zhao X: Role of hepatoma-derived growth factor in promoting *de novo* lipogenesis and tumorigenesis in hepatocellular carcinoma. Mol Oncol 12(9): 1480-1497, 2018. DOI: 10.1002/1878-0261.12357
24 Cheng C, Tai M, Chang H, Chou M, Chen G, Chen P, Kao Y: Oxidized low-density lipoprotein induced hepatoma-derived growth factor upregulation mediates foam cell formation of cultured rat aortic vascular smooth muscle cells. Eur J Cell Biol 100(5-6): 151169, 2021. DOI: 10.1016/j.ejcb.2021.151169
25 Gu W, Shen H, Xie L, Zhang X, Yang J: The role of feedback loops in targeted therapy for pancreatic cancer. Front Oncol 12: 800140, 2022. DOI: 10.3389/fonc.2022.800140
26 Masuzaki R, Kanda T, Sasaki R, Matsumoto N, Nirei K, Ogawa M, Karp SJ, Moriyama M, Kogure H: Suppressors of cytokine signaling and hepatocellular carcinoma. Cancers (Basel) 14(10): 2549, 2022. DOI: 10.3390/cancers14102549

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